A meta-analysis of prospective studies of coffee consumption and mortality for all causes, cancers and cardiovascular diseases.

Several prospective studies considered the relation between coffee consumption and mortality. Most studies, however, were underpowered to detect an association, since they included relatively few deaths. To obtain quantitative overall estimates, we combined all published data from prospective studies on the relation of coffee with mortality for all causes, all cancers, cardiovascular disease (CVD), coronary/ischemic heart disease (CHD/IHD) and stroke. A bibliography search, updated to January 2013, was carried out in PubMed and Embase to identify prospective observational studies providing quantitative estimates on mortality from all causes, cancer, CVD, CHD/IHD or stroke in relation to coffee consumption. A systematic review and meta-analysis was conducted to estimate overall relative risks (RR) and 95 % confidence intervals (CI) using random-effects models. The pooled RRs of all cause mortality for the study-specific highest versus low (≤1 cup/day) coffee drinking categories were 0.88 (95 % CI 0.84-0.93) based on all the 23 studies, and 0.87 (95 % CI 0.82-0.93) for the 19 smoking adjusting studies. The combined RRs for CVD mortality were 0.89 (95 % CI 0.77-1.02, 17 smoking adjusting studies) for the highest versus low drinking and 0.98 (95 % CI 0.95-1.00, 16 studies) for the increment of 1 cup/day. Compared with low drinking, the RRs for the highest consumption of coffee were 0.95 (95 % CI 0.78-1.15, 12 smoking adjusting studies) for CHD/IHD, 0.95 (95 % CI 0.70-1.29, 6 studies) for stroke, and 1.03 (95 % CI 0.97-1.10, 10 studies) for all cancers. This meta-analysis provides quantitative evidence that coffee intake is inversely related to all cause and, probably, CVD mortality.